

Stochastic Gradient Descent

A first motivation

Everything so far: $\nabla f(x)$, or $g_x \in \partial f(x)$

In this next sequence:

1. May not have full/exact access to $g_x \in \partial f(x)$
 - may only have access to a noisy version of g_x .
2. May have access to $g_x / \nabla f(x)$, but may be v. expensive to compute, so we instead use "approximate or noisy" - stochastic version.

Stochastic Gradient Descent - 3 main settings

Generic noisy gradients

⑧ Stochastic Optimization

Random coordinate descent

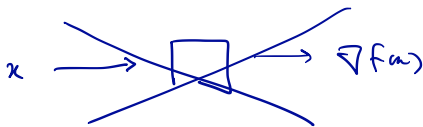
$$\begin{aligned} \min: & f(x) \\ \text{st: } & x \in \mathcal{X} \end{aligned}$$

$$x \rightarrow \boxed{} \rightarrow \begin{matrix} \nabla f(x) \\ g_x \in \partial f(x) \end{matrix}$$

instead :

$$x \rightarrow \boxed{} \rightarrow \tilde{g}_x$$
$$\mathbb{E}[\tilde{g}_x] \in \partial f(x).$$

Noisy Gradients



$$x \longrightarrow \square \longrightarrow \tilde{g}_x = \nabla f(x) + w$$

w is zero-mean noise.

$$\Rightarrow \mathbb{E}[\tilde{g}_x] = \mathbb{E}[\nabla f(x) + w] = \nabla f(x) + \cancel{\mathbb{E}[w]}^0$$

Stochastic Optimization

$$\begin{array}{ll} \min_x: f(x) & // \\ \text{st: } x \in \mathcal{X} & \end{array} \quad \begin{array}{ll} \min_x: \mathbb{E}_{\mathcal{Z}} [f(x; \mathcal{Z})] \\ \text{st: } x \in \mathcal{X} & \end{array}$$

In ML applications - supervised learning -
we have precisely this setting -

Ex: Regression problem: We want to learn how to predict
 y given $\mathcal{Z}$ - a vector of features.

If f is our loss function

then: $\mathbb{E}[f]$ - our expected loss.

We see: $\mathcal{Z}$ we predict: $\mathcal{Z}^T x$, truth: y
incur loss: $(y - \mathcal{Z}^T x)^2$.

$$\min_x: \mathbb{E}_{\mathcal{Z}} \left[(y - \mathcal{Z}^T x)^2 \right].$$

Stochastic Optimization :

Regression: $\min_x \mathbb{E}_{\tilde{z}} [(y - \tilde{z}^T x)^2]$

Common setting: seen "training data" and we want to use them to learn a good x .

$$\Rightarrow (y_1, \tilde{z}_1), (y_2, \tilde{z}_2), \dots, (y_n, \tilde{z}_n)$$

Empirical loss: $\frac{1}{n} \sum (y_i - \tilde{z}_i^T x)^2$. (ERM)

minimizing empirical loss: Empirical Risk Minimization

More generally :

$$\begin{cases} \min: f(x) = \frac{1}{n} \sum f_i(x) \\ \text{st: } x \in \mathcal{X} \end{cases}$$

(in regression setting: $f_i(x) = (y_i - \tilde{z}_i^T x)^2$.)

Stochastic Optimization

$$\min: \frac{1}{n} \sum f_i(x)$$

$$\text{st: } x \in \mathcal{X}$$

$$\begin{aligned} \text{GD: } x_{t+1} &= x_t - \eta \nabla \left(\frac{1}{n} \sum f_i(x) \right) \\ &= x_t - \eta \frac{1}{n} \sum_{i=1}^n \nabla f_i(x) \end{aligned}$$

If n is large (ERM: $n = \# \text{ data points}$)

then computing $\nabla f = \frac{1}{n} \sum \nabla f_i$

could be very expensive; requires a full pass over data.

Stochastic Optimization

$$f = \frac{1}{n} \sum f_i.$$

$$\nabla f = \frac{1}{n} \sum \nabla f_i.$$

$$x \xrightarrow{\square} \cancel{\nabla f} \quad \tilde{g}_x = \nabla f_I(x)$$

$I \sim \text{uniform}(1, \dots, n).$

Q: Is $\nabla f_I(x)$ a stochastic gradient?

$$\begin{aligned} \mathbb{E}[\nabla f_I(x)] &= \sum_{i=1}^n \nabla f_i(x) \cdot \frac{1}{n} = \frac{1}{n} \sum \nabla f_i(x) \\ &= \nabla f(x). \end{aligned}$$

Random coordinate descent

$$\begin{array}{ll} \min: & f(x) \\ \text{st}: & x \in \mathcal{X} \end{array} \quad \nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{pmatrix}, \quad x \in \mathbb{R}^d.$$

$$x \xrightarrow{\square} \cancel{\nabla f(x)} \quad \tilde{g}_x = d \begin{bmatrix} 0 \\ \vdots \\ \frac{\partial f}{\partial x_J} \\ \vdots \\ 0 \end{bmatrix}, \quad J \sim \text{uni}\{1, \dots, d\}$$

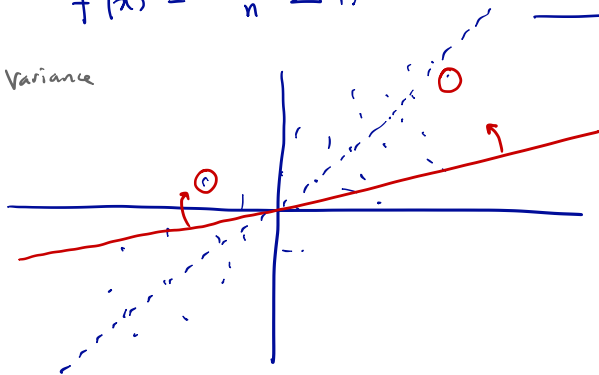
$$\mathbb{E}_J [\tilde{g}_x] = \sum_{j=1}^d A \begin{bmatrix} 0 \\ \vdots \\ \frac{\partial f}{\partial x_j} \\ \vdots \\ 0 \end{bmatrix} \cdot \frac{1}{d}$$

$$= \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{bmatrix} = \nabla f(x)$$

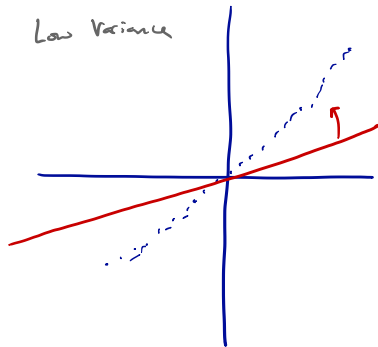
SGD - the picture (role of variance)

$$f(x) = \frac{1}{n} \sum f_i(x) \quad \underline{\text{ERM.}}$$

High Variance



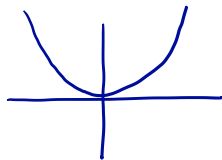
Low Variance



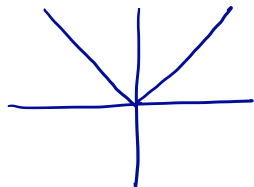
$$\text{SGD: } x_{t+1} = x_t - \eta \tilde{g}_x$$

Connection to past results: self tuning?

GD and sub-GD :



(A)



(B)

(A) "self-tuning"

As $x \rightarrow x^*$,
 $\nabla f(x) \rightarrow 0$

(B) No self-tuning

$\Rightarrow$ forced to take
small step size

$\Rightarrow$ slower convergence.